MIL-D-48025A (PA)
14 March 1975
SUPERSEDING
MIL-D-48025 (MU)
01 February 1972

MILITARY SPECIFICATION

DETONATOR, STAB, M99
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing for one type of stab detonator designated as Detonator, Stab, M99.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-650 - Explosives: Sampling Inspection and Testing
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

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DRAWINGS

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* PRODUCT AND PACKING DRAWINGS

9272235 - Detonator, Stab, M99
9258182 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

* INSPECTION EQUIPMENT DRAWINGS

9200754 - Firing Assembly
9200012 - Firing Pin
8816367 - Firing Pin Guide
8855295 - Lead, Coined Disc
8827829 - Bushing
8868629 - Steel Ball

* (For meaning of asterisk (*) see 6.5).

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator. - The detonator shall comply with all requirements specified on Drawing (dwg.) 9272235, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. - The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent when tested.

3.4 Moisture content of primer mixture. - The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent when tested.

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3.5 Functioning

3.5.1 Output. - The detonator shall function on the first attempt and shall cause perforation of a coined lead disc to a diameter of not less than 3/16 inch when tested.

*3.6 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.7 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First article inspection
b. Quality conformance inspection

*4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of fifty (50) complete sets of parts and subassemblies (i.e., 50 each of every component and sub-assembly. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2, testing per para. 4.4.3, and Table I specified herein.

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TABLE I
FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48025A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*	Detonator, Stab, M99, Inert Parts				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Parts (Dwg. 9272235) Examination for Defects	50(a)		3.2	4.4.2.1 thru 4.4.2.5
	NOTE: 50 each of every component and sub-assembly.				

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4.3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts of one lot interfix number from one supplier.
b. Primer mixture from one batch or blending of batches which results in homogeneity of all igniter batches used in loading the entire detonator lot.
c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION MIL-D-48025A (PA)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Disc, Detonator				9272237
					NEXT HIGHER ASSEMBLY 9272235
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Diameter		0.40%	3.2	Gage
132.	Thickness		0.40%	3.2	Gage
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.2	Cup, Detonator		9272236
			NEXT HIGHER ASSEMBLY 9272235

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Diameter of hole, min.		0.40%	3.2	Gage
132.	Concentricity of flash hole with outside diameter		0.40%	3.2	Gage
133.	Length		0.40%	3.2	Gage
134.	Outside diameter		0.40%	3.2	Gage
Minor:					
201.	Inside diameter		0.65%	3.2	Gage
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.3	Retainer				9272239
					NEXT HIGHER ASSEMBLY 9272235
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131:	Outside diameter, min.		0.40%	3.2	Gage
132.	Total length, min.		0.40%	3.2	Gage
133.	Thickness through bottom, min		0.40%	3.2	Gage
134.	Profile of point, min.		0.40%	3.2	Gage
Minor:					
201.	Outside diameter, max		0.65%	3.2	Gage
202.	Total length, max.		0.65%	3.2	Gage
203.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.4	Assembly, Prior to Loading				9272235
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Coating compound or lacquer on exposed surface of disc		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.5	Assembly				9272235
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Cup cracked or split		100%	3.2	Visual
2.	Retainer cracked or split		100%	3.2	Visual
Major B:					
131.	Length		0.40%	3.2	Gage
132.	Diameter		0.40%	3.2	Gage
133.	Disc, loose, punctured, torn or wrinkled to extent that content s are exposed		0.40%	3.2	Visual
134.	Color of lacquer incorrect		0.40%	3.2	Visual
135.	Cup damaged except as described in critical above		0.40%	3.2	Visual
136.	Crimp not full 360 degrees		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.6	Box, Set-up, Inner, After Loading and Prior to Closing Slide Cover		9258182
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Filler and cushion assembly missing		100%	3.2	Visual
2.	Filler and cushion assembly improperly assembled		100%	3.2	Visual
3.	Stiffner missing		100%	3.2	Visual
4.	Slide cover assembled reversed		100%	3.2	Visual
Major A:					
101.	Release paper missing		0.40%	3.2	Visual
Minor:					
201.	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.7	Box, Set-up, Outer, with Inner Set-up Boxes Assembled and Prior to Assembling Top				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Red surface on inner set-up boxes do not alternate on same side of outer set-up boxes		100%	3.2	Visual
2.	Stiffner not facing one direction		100%	3.2	Visual
Major A&B:	None defined				
Minor:					
201.	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.8	Sealed Packed Barrier Bag, Prior to Packing in Box		9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Watervaporproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.2	Visual
102.	Evidence of crushed box		0.40%	3.2	Visual
NOTE:	Package should be opened if there is evidence of crushing.				
Minor:					
201.	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202.	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-48025A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.9	Box, Packing, Ammunition Prior to Assembling Cover and Strapping				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Bags containing set-up boxes damaged		0.40%	3.2	Visual
102.	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.2	Visual
103.	Marking missing, misleading or unidentifiable		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48025A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.10	Box, Packing, Ammunition				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A:					
101.	Strapping missing, broken, loose or improperly fastened		0.40%	3.22	Visual/Manual
102.	Box damaged		0.40%	3.2	Visual
103.	Marking missing, incomplete or illegible		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.6	Visual

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4.4.3 Testing

4.4.3.1 Chemical sampling. - A representative sample shall be randomly selected for the chemical tests specified in 4.5. The procedure described in ASTM E300 should be used to withdraw the samples. The tests in 4.5.1, 4.5.2 and 4.5.4 shall be performed using prescribed analytical procedure for replicate determinations given in standard analytical textbooks.

4.4.3.2 Moisture

*4.4.3.2.1 Moisture content of lead azide (See 3.3). - Major A Defect - The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample shall be selected and tested as specified in 4.5.1. If the moisture content is in excess of the percentage specified and loading has not yet begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

*4.4.3.2.2 Moisture content of primer mixture (See 3.4). - Major A defect - The moisture content of the primer mixture shall be determined for each lot at the time of loading. A sample shall be selected and tested as specified in 4.5.2. If the moisture content is in excess of the percentgae specified and loading has not yet begun, the lot of primer mix shall be rejected. If loading has begun, action on the unloaded primer mix shall be as specified above and all detonators loaded with the questionable primer mix shall be rejected.

4.4.3.3 Functioning

*4.4.3.3.1 Output (See 3.5.1) - Major A defect - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If one or two defects are found, a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample a total of 4 or more defects are found, the lot shall be rejected. The test shall be performed as specified in 4.5.3.1 using test equipment as specified in 4.4.4.

*4.4.3.4 Composition of primer mixture (see dwg. 9272235). - Major A defect - The composition of each lot of primer mixture shall be determined at the time of loading. A sampling of suffi-

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cient size to perform the test specified in 4.5.4 shall be selected for the determination. If the composition of primer mixture is in excess of the specified percentage and the loading has not begun, the lot shall be rejected. If detonator loading is in progress, the remaining unloaded primer mixture in the lot with all the detonators loaded with the nonconforming primer mixture shall be rejected.

*4.4.3.5 Heat seal test of sealed bag (see dwg. 9258183). - Major A defect - Sampling, acceptance criteria, and test procedures for this test shall be as specified in MIL-P-116.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

4.5 Test methods and procedures

4.5.1 Moisture content of lead azide. - The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene when weighing.

4.5.2 Moisture content of the primer mixture. - The moisture content of the primer mixture shall be determined as specified in 4.5.1 except that the drying temperature shall be 60 plus or minus 5 degrees C. and only 0.40 to 0.70 grams of primer mixture shall be dried.

4.5.3 Functioning

4.5.3.1 Output. - The detonator shall be assembled to the applicable test fixture specified in 4.4.4. A steel ball 3/4 inch in diameter and weighing 1 ounce, max., shall be dropped on the firing pin from a height of 6 inches, max. The height of the drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Method of inspection shall be in accordance with dwg. 8816367, 8827829, 8855295, 8868629, 9200012 and 9200754.

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4.5.4 Composition of primer mixture NOL #130

4.5.4.1 Barium nitrate. - An accurately weighed portion of approximately 0.5 grams of primer mixture shall be transferred to a tared medium porosity, fritted glass crucible of 20 ml. capacity. Three ml. of lead azide saturated distilled water shall be added at a temperature of 50 degree C., plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3 ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percentage} = \frac{(A-B)\ 100}{W(1-.01M)}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.5.2 in percent.
W = weight of sample

4.5.4.2 Basic lead styphnate (when applicable). - The residue obtained in the determination of barium nitrate shall be extracted with six 5 ml. portions of saturated ammonium acetate solution at about 25 degrees C. (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for 2 minutes.
The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250 ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml. aliquot of this solution shall be transferred to a 50 ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm.) and its optical density determined at a wave length of 410 milli-

96%

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microns. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated as a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\,(A-B)}{W(1.-01M)D}$$

where:

A = optical density of basic lead styphnate solution
B = optical density of ammonium acetate solution
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.2 in percent.
W = weight of sample
F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer thelead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.4.3 "Determination of normal lead styphnate (when applicable). - Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated) at about 25°C. until the washings are colorless. Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination.

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Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its optical density at a wave length of 410 millimicrons. Use as reference, a similar cell containing ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the optical density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millimicrons the optical density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:"

$$\text{Normal lead styphnate, percent} = \frac{F(A-B)}{WD}$$

where:

A = optical density of normal lead styphnate solution
B = optical density of ammonium acetate solution
D = width of corex cell, cm.
W = weight of the dried sample
F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution to the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.4.4 Tetracene. - The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C. plus or minus 5 degrees

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C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C. plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.
W = weight of the sample
M = moisture content of sample obtained from 4.5.2 in percent.

4.5.4.5 Antimony sulfide. - The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W(1-.01M)}$$

where:

D = weight of crucible and sample after treating with hot water.
E = weight of crucible
W = weight of sample
M = moisture content of sample obtained in 4.5.2 in percent.

4.5.4.6 Lead azide. - The lead azide content of the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and antimony sulfide.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

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*5.1.1 Level C. - The detonators shall be packaged in accordance with dwg. 9258182 and dwg. 9258183.

*5.1.2 Level C. - Detonator parts - Packaging shall be in accordance with MIL-STD-1169 requirements.

5.2 Packing

*5.2.1 Level C. - Packing for detonators shall be as specified on dwg. 9258184.

*5.2.2 Level C. - Detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

*5.3 Marking. - Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require

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that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

6.4.1 Functioning. - A sample of 315 detonators shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defect is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 5 or more defects are found, the lot shall be rejected. The test shall be performed as specified in 4.5.3 using test equipment as specified in 4.4.4.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-123

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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